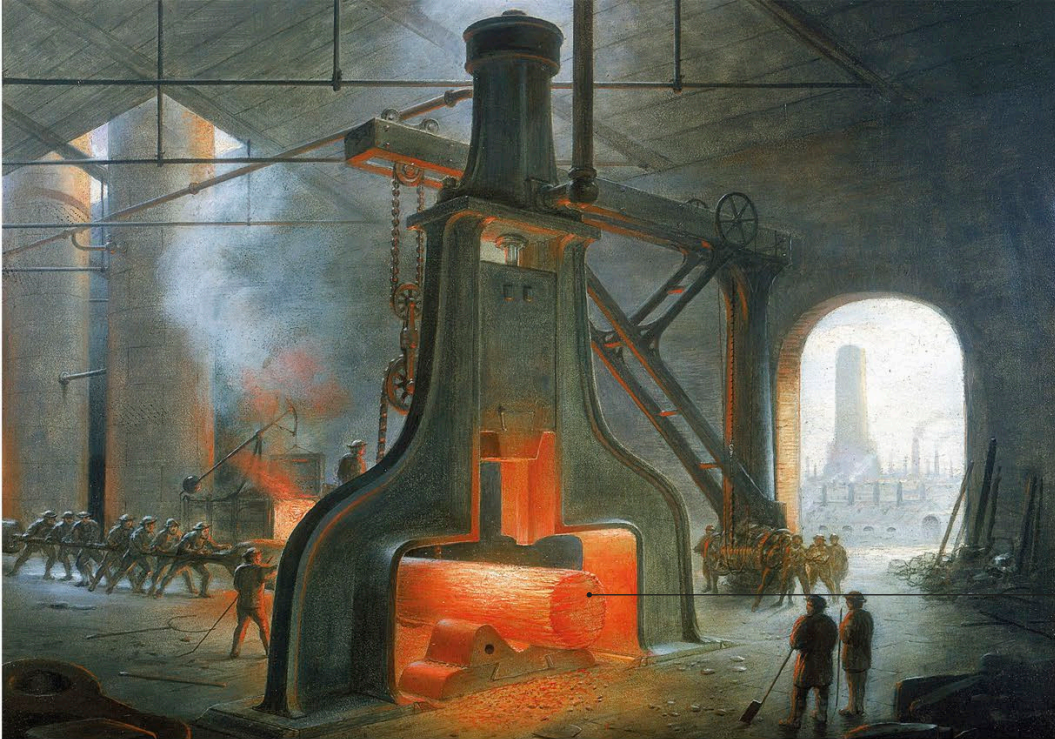




STEAM HAMMER

When English engineer Isambard Kingdom Brunel started work on his ship *SS Great Britain*, he discovered that hammering out the giant shafts for its paddle wheels was beyond human ability. Scottish engineer James Nasmyth came up with the idea of a giant steam-driven hammer. He made the first in 1840 and patented it in 1842.

Workers feed a red-hot piece of iron into a steam hammer, which pounds the metal into shape.

SPANNER FORCES

When someone rotates a spanner, they apply a twisting force called torque. If force is applied further from the point of rotation, torque is greater and the spanner is easier to turn.

Force is applied to the end of the spanner.



The further from the nut the force is applied, the greater the torque.

SPIRIT LEVEL

A bubble in a liquid always rises to the highest point. In an upward-curved level tube it will settle at the center. This was recognized by the French scientist Melchisédech Thévenot, who invented the first spirit level in 1661. Spirit levels have been used by builders ever since to ensure their work is perfectly horizontal or vertical.

A yellow-colored liquid is used for easy reading.



Modern spirit level giving vertical, horizontal, and angle readings



The jaw is adjusted by a rotating screw just under it.

ADJUSTABLE SPANNER

An adjustable spanner has a movable jaw, which allows it to be used with different-sized nuts and bolts. The English agricultural engineer Richard Clyburn is credited with its invention in 1842, while he was working at an iron works in Gloucester, UK.

The metals are melted so they can be bonded together on cooling.



ARC WELDING

Since ancient times, blacksmiths have used concentrated heat to bond metals together. In 1881, French inventor Auguste de Méritens invented a way of using electricity to create enough heat to melt metals that would then be joined when cooled, in a process known as arc welding.